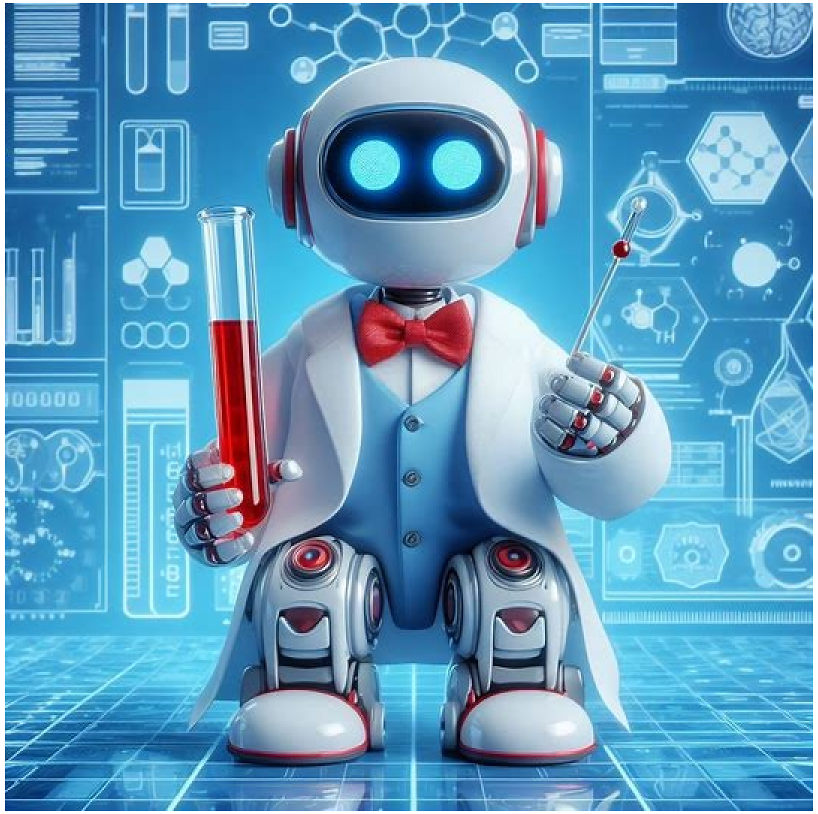


AI in Drug Discovery – An Overview

Andrea Volkamer and Pat Walters

September 2th, 2026

Session 2 - Data is all you need!

Who we are!

Pat Walters
OpenADMET

Nessa Carson
AstraZeneca

Ana Mirandela
Saarland University

Lisa-Marie Rolli
Saarland University

Andrea Volkamer
Saarland University

What we will do today

Session 1 - 1:30 - 2:30 pm

- An introduction to Artificial Intelligence (AI) and Machine Learning (ML)
- Molecular representations
- AI architectures

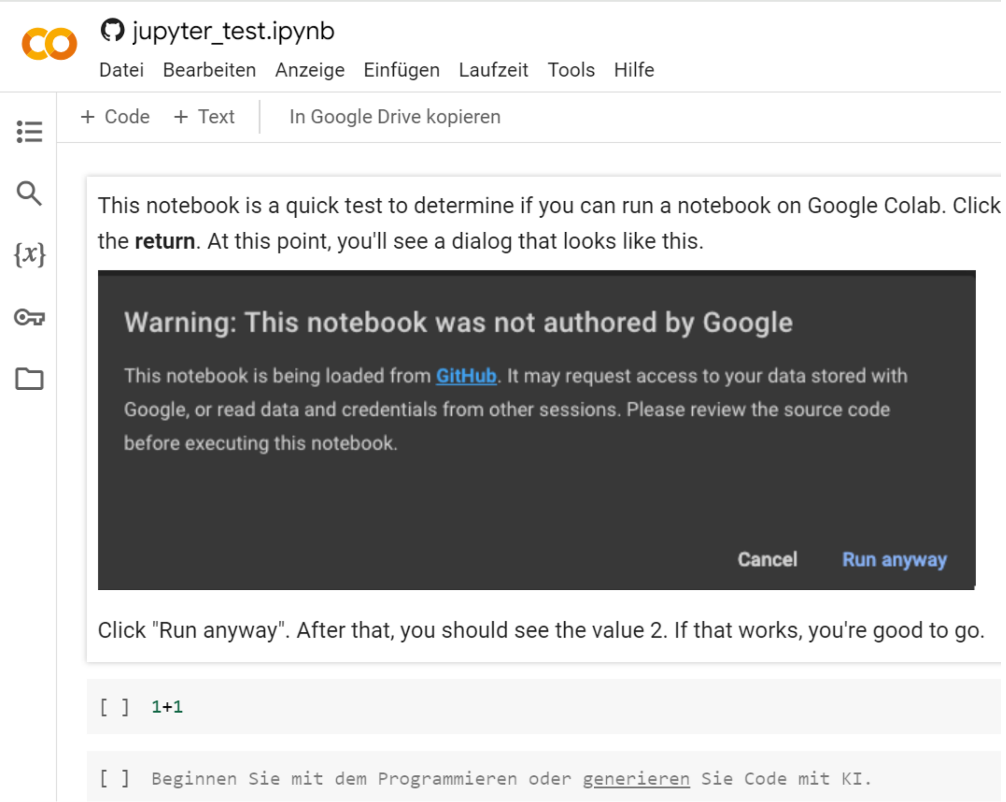
Session 2 - 3:00 - 4:00 pm

- Importance of data quality for AI and ML
- Data sets and (pre)processing
- Trusting your predictions

Session 3 - 4:30 - 5:30 pm

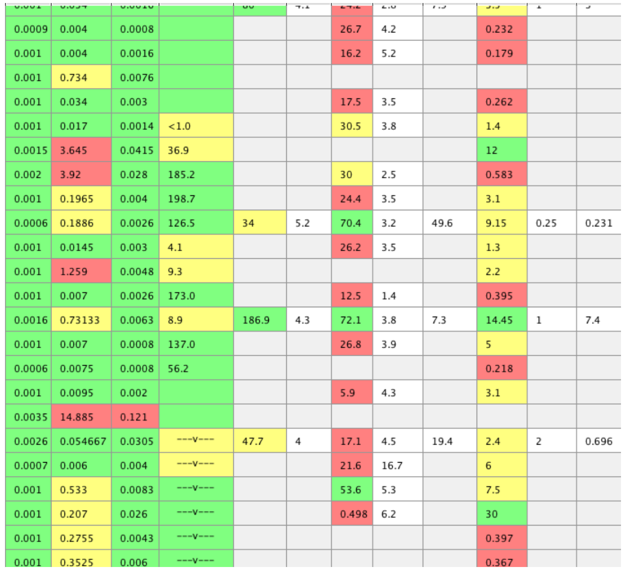
- AI in practice
- Molecule generation
- Active learning

Lectures supported by hands-on sessions …



Key Components of Machine Learning in Drug Discovery (or Anything Else)

Data



0.0009	0.004	0.0008				26.7	4.2		0.232		
0.001	0.004	0.0016				16.2	5.2		0.179		
0.001	0.734	0.0076									
0.001	0.034	0.003				17.5	3.5		0.262		
0.001	0.017	0.0014	<1.0			30.5	3.8		1.4		
0.0015	3.645	0.0415	36.9						12		
0.002	3.92	0.028	185.2			30	2.5		0.583		
0.001	0.1965	0.004	198.7			24.4	3.5		3.1		
0.0006	0.1886	0.0026	126.5	34	5.2	70.4	3.2	49.6	9.15	0.25	0.231
0.001	0.0145	0.003	4.1			26.2	3.5		1.3		
0.001	1.259	0.0048	9.3						2.2		
0.001	0.007	0.0026	173.0			12.5	1.4		0.395		
0.0016	0.73133	0.0063	8.9	186.9	4.3	72.1	3.8	7.3	14.45	1	7.4
0.001	0.007	0.0008	137.0			26.8	3.9		5		
0.0006	0.0075	0.0008	56.2						0.218		
0.001	0.0095	0.002				5.9	4.3		3.1		
0.0035	14.885	0.121									
0.0026	0.054667	0.0305	---v---	47.7	4	17.1	4.5	19.4	2.4	2	0.696
0.0007	0.006	0.004	---v---			21.6	16.7		6		
0.001	0.533	0.0083	---v---			53.6	5.3		7.5		
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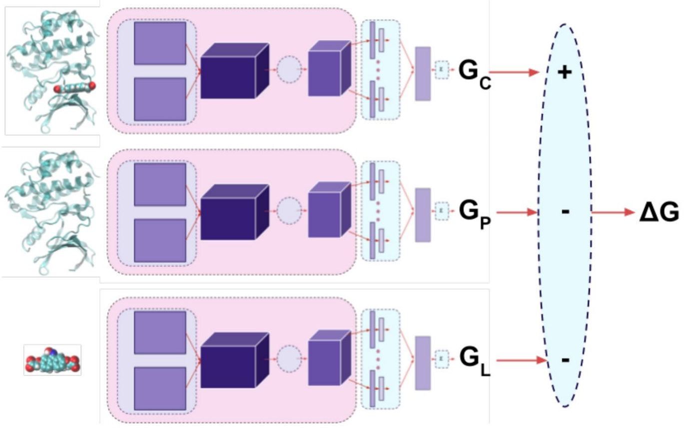
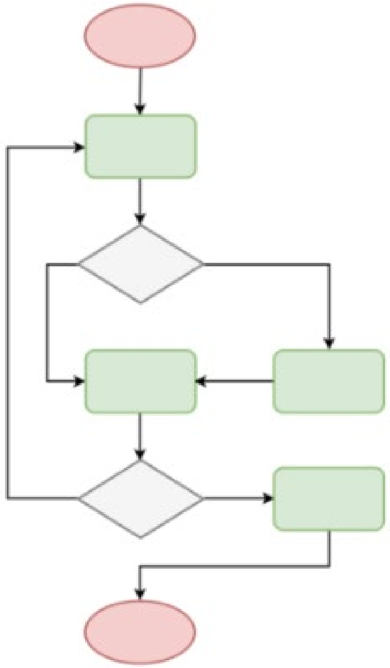
0.001	0.2755	0.0043	---v---						0.397		
0.001	0.3525	0.006	---v---						0.367		

Representation



Algorithms



AI is Full of Promise – Still Short on Proof

10+ yrs

of AI in drug R&D

Few

assets reach phase III

Why?

underspecified problems + conditional, biased data

Perspective | Published: 07 August 2026

Artificial intelligence in drug discovery – what it is, where we stand and the path forward

Andreas Bender, Morgan C. Thomas, Jack W. Scannell, David A. Shaywitz, Gian Marco Ghiandoni, Joe G. Greener, Lavinia-Lorena Pruteanu, Rachel DeVay Jacobson, Koichi Handa, Mariko Hirano, Srijit Seal, Manas Mahale, Marco F. Schmidt, Tim Ahfeldt, Francesca Grisoni & Isidro Cortes-Ciriano

Nature Reviews Drug Discovery (2026) | Cite this article

Artificial intelligence (AI) in drug discovery has attracted increasing interest over the past decade. It is now time for a critical review of progress in the field: where did we advance – and where are we yet to see impact – when it comes to what matters in drug discovery, which is to deliver safer and more efficacious medicines to patients faster? Although a wide variety of AI methods have been developed, applied and benchmarked, evidence of their clinically relevant impact is, so far, disappointingly limited. In this Perspective we discuss potential reasons, including an insufficient focus on clinical translation during model development,

Where Machine Learning Excels ⟷ Pharmaceutical Data Properties

(Ultra)Large amounts of data
- Pharmaceutical data is miniscule compared to many other fields

Responses are definitive

Samples are independently distributed
- No cases where the same example falls into 2 different categories

Samples are identically distributed
- Equal distributions of positive and negative examples

Training data is representative of what is being predicted

Data is sparse
- Rarely have a complete data matrix

Data is truncated
- Many assay values report as “<1” or “>30”
- Difficult to know the true value

Data has a limited dynamic range
- Often spans only 2 or 3 logs
- Small dynamic range combined with experimental error makes significant correlations difficult

Even data from the “same” assay can be heterogeneous
- PK data measured with different doses, formulations
- Response can vary with operator, equipment, lab

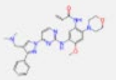
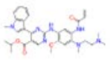
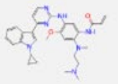
Data covers a limited chemical space
- Even global models can be local

Which data sets are used in the domain?

Which can we recommend?

Small Molecule Bioactivity Datasets: ChEMBL or PubChem

Database for collecting binding affinities

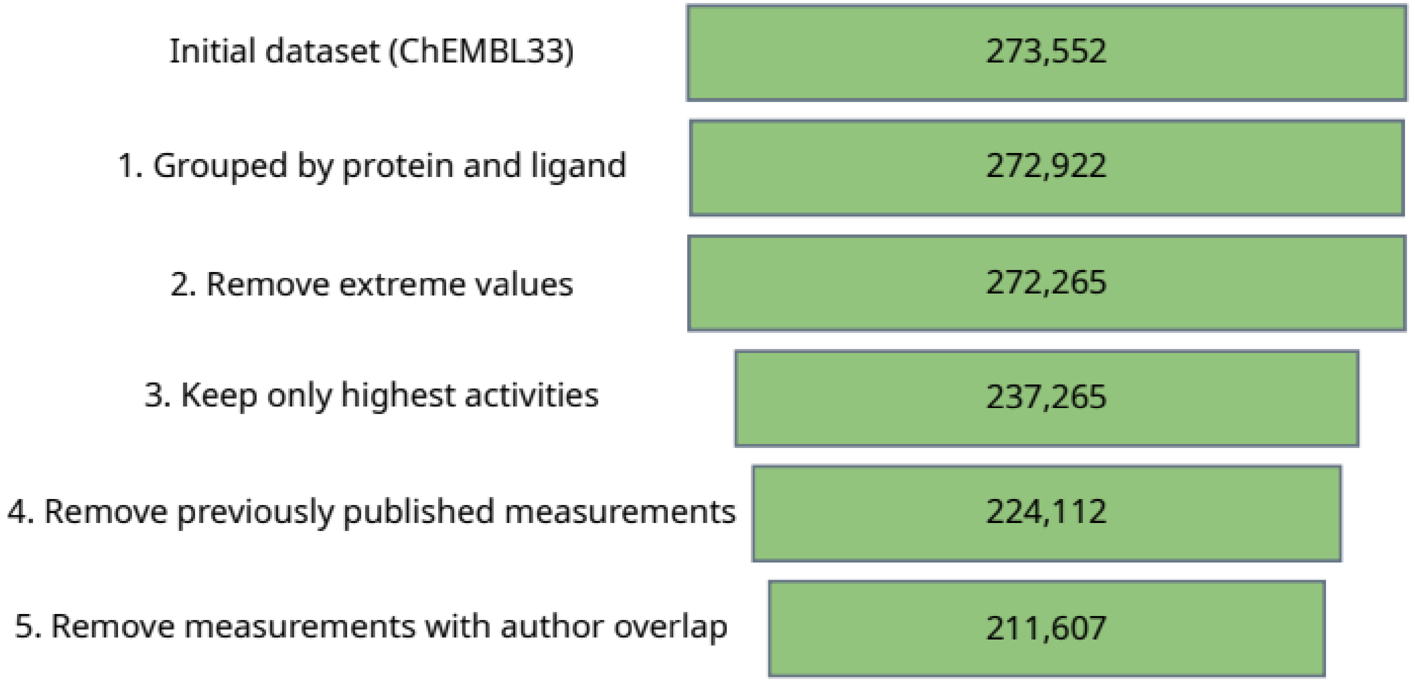
ChEMBL ID	Search Hit	Name	Synonyms	Type	Max Phase	Molecular Weight	Targets	Bioactivities
 CHEMBL4558324		LAZERTINIB	C-18112003-G, GNS1480, GNS-1480, JNJ-73841937-AAA, Lazertinib, Yh25448, YH25448, YH-25448	Small molecule	3	554.66	12 By Type: 	44 By Std. Type: 
 CHEMBL4650319		MOBOCERTINIB	AP32788, AP-32788, Mobocertinib, Tak-788, TAK-788	Small molecule	4	585.71	14 By Type: 	55 By Std. Type: 
 CHEMBL4761468		AUMOLERTINIB	Almonertinib, Ameile, Amerol, Aumolertinib, Egfr t790m inhibitor hs-10296, EQ143, EQ-143, HS-10206, Hs 10296, Hs-10296, HS-10296	Small molecule	3	525.66	5 By Type: 	23 By Std. Type: 

https://www.ebi.ac.uk/chembl/

How to Preprocess such Data Properly?

- Automated curation pipeline to reduce errors and to increase reliability

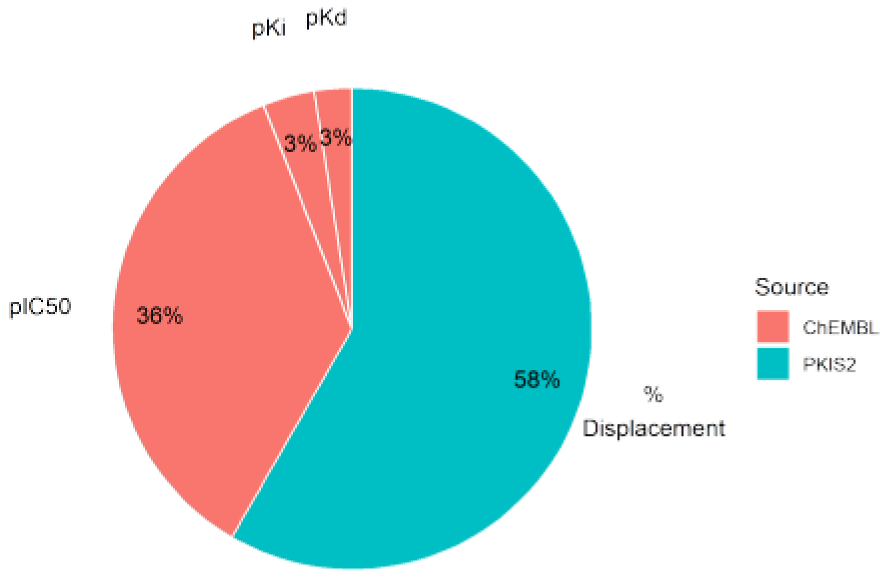
  see Kramer, et al. *JMedChem* **2012**; 55(11):5165–5173

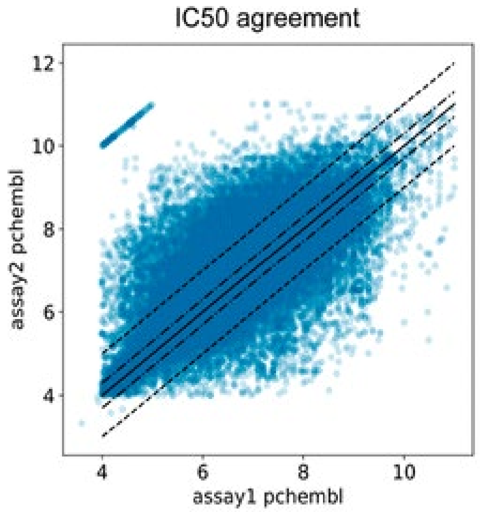


López-Ríos de Castro, *et al.*, *biorxiv*, **2024**
https://doi.org/10.1101/2024.09.10.612176

How to Preprocess such Data Properly?

- Bioactivity assay measurement classes vary by and within data set





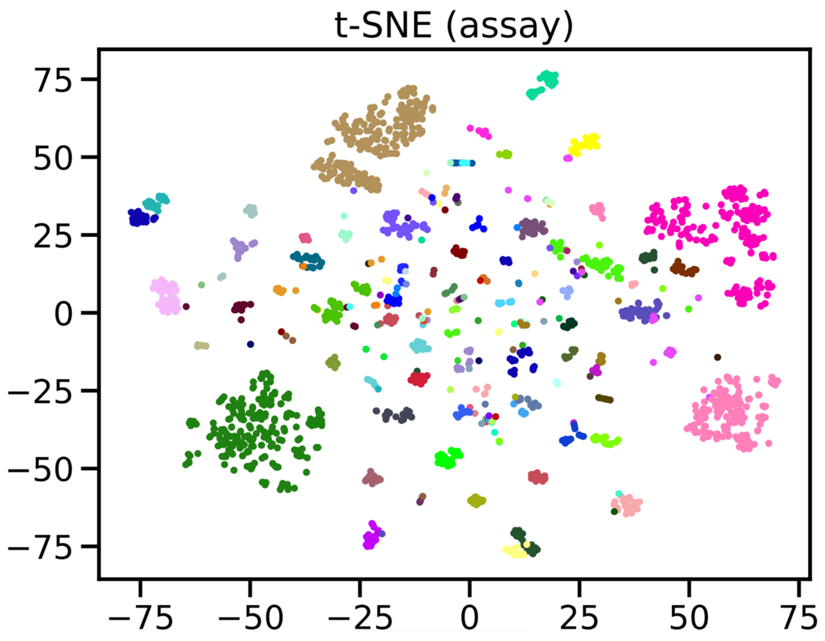
R2 = 0.31
MAE = 0.50
Kendall's tau = 0.51

López-Ríos de Castro, *et al.*, *biorxiv*, **2024**
https://doi.org/10.1101/2024.09.10.612176

Landrum, Riniker., JCIM, **2024**, 64,5
https://doi.org/10.1101/2024.09.10.612176

Typical Challenges in (Public) Chemical Datasets

- Different experimental protocols
- Inconsistent values
- Missing documentation
- Data tends to be heavily clustered



- kinodata-EGFR ligand activities
- t-SNE on 2048 bit Morgan fingerprints
- colors = assays

Standardization: One Molecule, One Representation

The same compound can appear in many different representations

- Different SMILES / atom ordering
- Salts, solvents & counterions
- Different protonation states
- Tautomers
- Stereochemistry inconsistencies
- Duplicates and conflicting records

Why does this matter for ML?

- Inconsistent representations → inconsistent features → unreliable comparisons
- A model may appear to perform differently simply because the input chemistry was not standardized consistently.

➢ Reproducible, application-aware standardization pipeline(s)

https://github.com/basf/Molpipeline

Some Literature Datasets are More Problematic Than Others

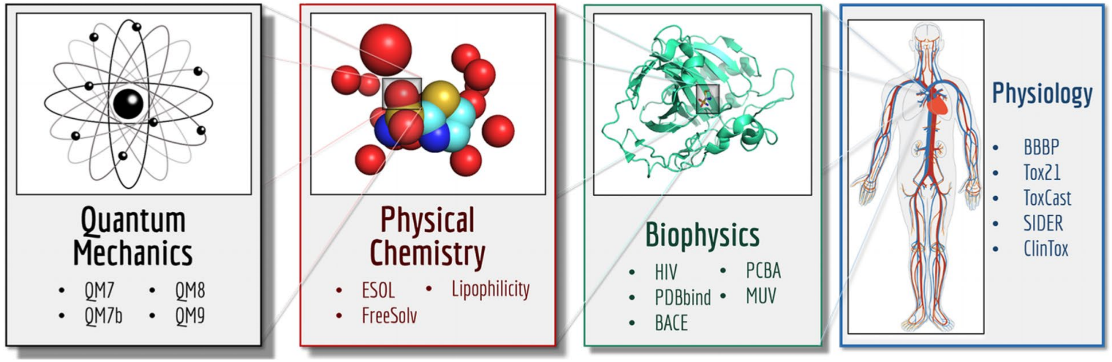

Check for updates
comment

Artificial intelligence foundation for therapeutic science

Artificial intelligence (AI) is poised to transform therapeutic science. Therapeutics Data Commons is an initiative to access and evaluate AI capability across therapeutic modalities and stages of discovery, establishing a foundation for understanding which AI methods are most suitable and why.

Kexin Huang, Tianfan Fu, Wenhao Gao, Yue Zhao, Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun and Marinka Zitnik

Therapeutic Data Commons



MoleculeNet

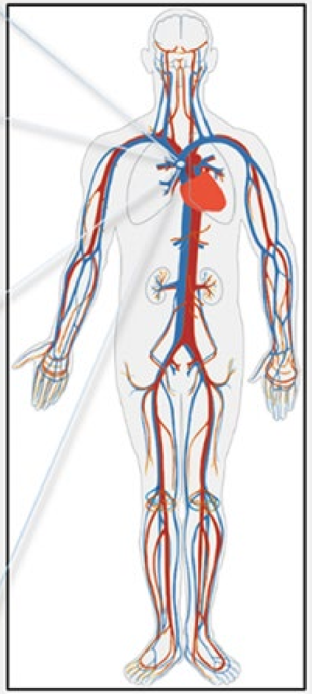
Some issues with

- Chemical structure errors
- Unspecified stereochemistry
- Inconsistent experiments
- Curation errors
- Unrealistic dynamic range
- Irrelevant experiments
- Poorly defined endpoints
- Assay artifacts

Please Don't Use These Datasets!

http://practicalcheminformatics.blogspot.com/2023/08/we-need-better-benchmarks-for-machine.html

For Now, Focus on Simple, Consistent, Well-Defined Endpoints



Physiology

- BBBP
- Tox21
- ToxCast
- SIDER
- ClinTox

Hepatobiliary disorders
Metabolism and nutrition disorders
Product issues
Eye disorders
Investigations
Musculoskeletal and connective tissue disorders
Gastrointestinal disorders
Social circumstances
Immune system disorders
Reproductive system and breast disorders
Neoplasms benign, malignant and unspecified (incl cysts and polyps)
General disorders and administration site conditions
Endocrine disorders
Surgical and medical procedures
Vascular disorders
Blood and lymphatic system disorders
Skin and subcutaneous tissue disorders
Congenital, familial and genetic disorders
Infections and infestations
Respiratory, thoracic and mediastinal disorders
Psychiatric disorders
Renal and urinary disorders
Pregnancy, puerperium and perinatal conditions
Ear and labyrinth disorders
Cardiac disorders
Nervous system disorders
Injury, poisoning and procedural complications

Recommended Datasets Supplied By the Polaris Initiative

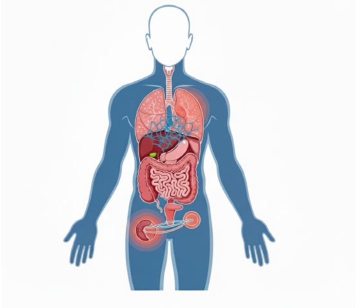
Polaris certified datasets have been evaluated and approved by industry experts

Dataset	Tags	Details
asap-discovery/antiviral-admet-2025-unblinded	V2, Molecule, ADMET	Size: 560; 2025-03-28
asap-discovery/antiviral-potency-2025-unblin...	V2, Molecule, Potency	Size: 1,328; 2025-03-28
asap-discovery/antiviral-ligand-poses-202...	V2, Molecule, Molecule 3 D	Size: 965; 2025-03-28
recursion/rxrx3-core	V2, Molecule, Image	Size: 222,601; 2024-12-11
roman-bushuiev/massspecgym	V2, Molecule, Small molecule discovery	Size: 231,104; 2024-11-26
leash-bio/belka-v1	V2, Molecule, Screen	Size: ~99.3M; 2024-10-30
adaptyv-bio/egfr-binders-v1	protein-design	Size: 213; 66 KB; 1 benchmark; 2024-10-25
adaptyv-bio/egfr-binders-v0	protein-design	Size: 202; 61 KB; 1 benchmark; 2024-09-26
polaris/drewry2017-pkis2-subset-v2	Kinase, HitDiscovery	Size: 640; 54 KB; 14 benchmarks; 2024-07-10
biogen/adme-fang-v1	adme	Size: 3,521; 384 KB; 7 benchmarks; 2024-07-10

https://polarishub.io/datasets?certifiedOnly=true

Complimentary Efforts That Can Address Some of the Gaps

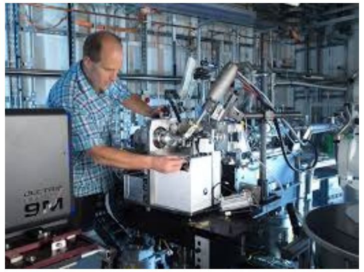
OpenADMET



How drugs interact with the body

- Absorption
- Metabolism
- Excretion
- Toxicology

OpenBind



How drugs bind to proteins

- Structure
- Binding affinity
- 500K structures / 5yrs

Data Generation and Dissemination

Blind Challenges to Test Methods

Establishing Best Practices

CBI CENTER FOR BIOINFORMATICS

So can I Trust my Model / Prediction

Data

[illegible]	[illegible]	[illegible]		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
0.0009	0.004	0.0008				26.7	4.2		0.232		
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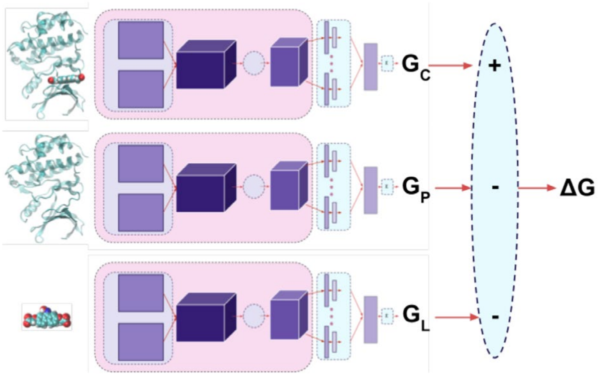
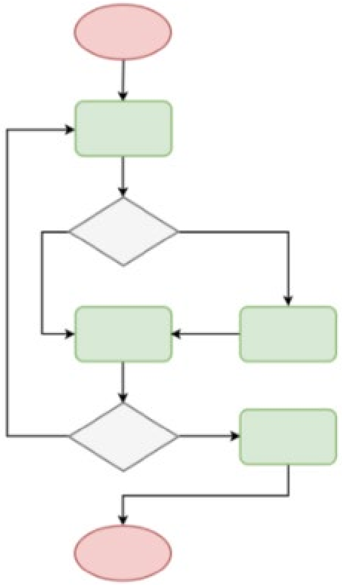
0.001	0.2755	0.0043	---v---						0.397		
0.001	0.3525	0.006	---v---						0.367		

Representation



Algorithms



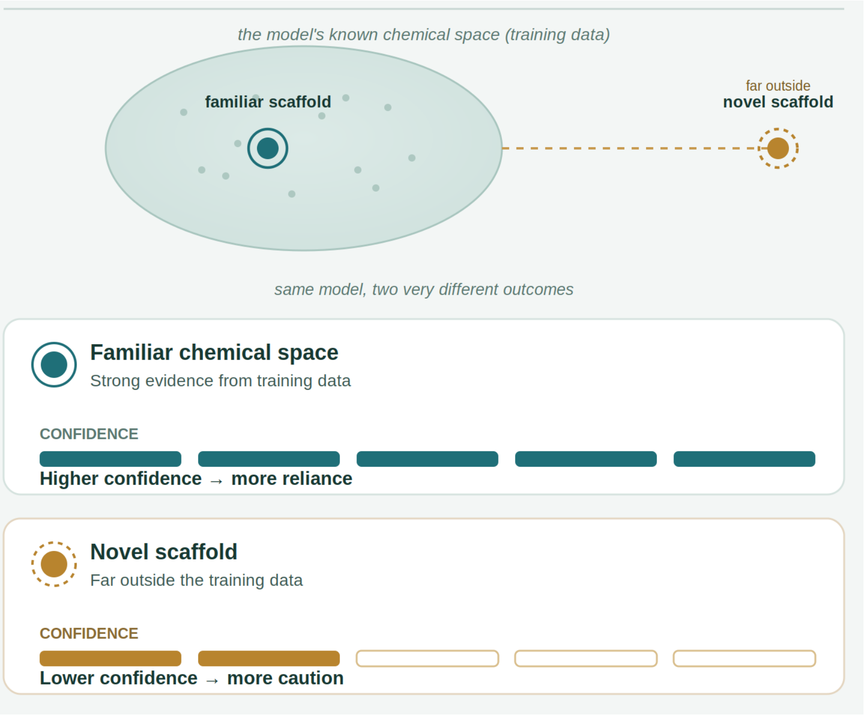
Applying ML to Drug Design with Trust

- **Trust:** *What the chemist believes about the model prediction*
  → "I expect this AI model to help me identify promising compounds."

- **Trustworthiness:** *What the model actually warrants*
  → "The model is worth to be trusted for this target, chemical space and intended use."

- **Calibrated trust:** *The match between the two*
  → "I trust on the model in proportion to how trustworthy its prediction is in this situation."

Virtual Screening
100,000 compounds are ranked by an ML model
Which selects 50 promising candidates



How do we Earn *Calibrated* Trust in an AI Model?

Trustworthiness should be built from multiple building blocks,
not a single benchmark score

Trustworthiness property	**Definition**	**Evidence / methods**
Performance: **Does it work on its task?**	Ability of a model or algorithm to achieve its intended task	Data quality & provenance · Appropriate splits/independet validation · Metrics
Reliability: **Does it work in domain?**	Ability to function correctly on data drawn from the same distribution*	Calibration · Ensembles · Conformal prediction · Applicability domain
Robustness: **Does it work out of domain?**	Ability to function correctly on data drawn from another distribution*	Scaffold / temporal splits · Stress testing · External validation
Interpretability: **How understandable is it?**	Extent to which a human can understand the decisions of a model	Feature / attribution analysis · Explainability · Failure cases

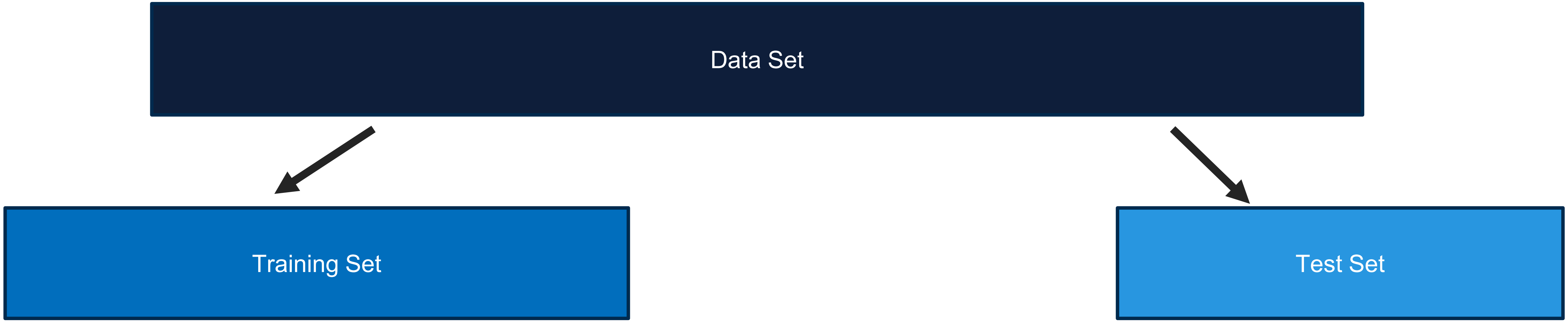
*as/than the training data

Increasing trust through

Performance

Does the model work on ist task?

How to Measure Performance?

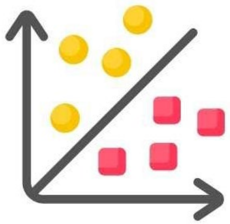


Training and **test** set are pairwise disjoint.

- Model is **fit** on the training set
- Model is **evaluated** the performance on unseen (test) data.

Performance Metrics Depends on the Prediction Task

Classification



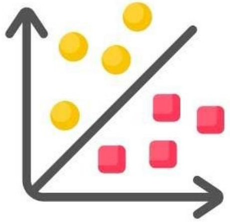
- Accuracy, sensitivity, specificity
- Area Under the Reciever Operator Curve (AUROC)
- Area Under the Precision Recall Curve (AUPRC)
- **Matthews Correlation Coefficient (MCC)**
- $\boldsymbol{F_1}$ **score**

Regression

- Mean Square Error (MSE)
- Pearson Correlation Coefficient (PCC)
- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of determination (R^2)

In all cases, compare your model to a simple baseline / null model !!!

Evaluation of Classifier on Test Data: Matthews Correlation Coefficient (MCC)



Confusion Matrix

	Measured positive	Measured negative
predicted positive	TP	FP
predicted negative	FN	TN

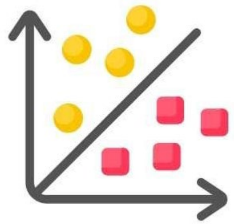
$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)\cdot(TP+FN)\cdot(TN+FP)\cdot(TN+FN)}} \in [-1,1]$$

- –1 means there is no correct prediction
- 0 indicates random prediction
- 1 means all predictions are correct

MCC is the binary version of Pearson correlation coefficient

+ Comparable across different models (because of the fixed range)
+ Uses all four confusion matrix components
+ Good for imbalanced data

Evaluation of Classifier on Test Data: F_1 Score



Confusion Matrix

	Measured positive	Measured negative
predicted positive	TP	FP
predicted negative	FN	TN

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Precision: $\frac{TP}{TP+FP}$

Recall: $\frac{TP}{TP+FN}$

- ➢ 1 perfect precision and recall
- ➢ 0 worst performance

\+ Simple and useful when precision and recall of the positive class are both important

\+ Good for imbalanced data

\+ Ignores true negatives

Evaluation of Regressor on Test Data

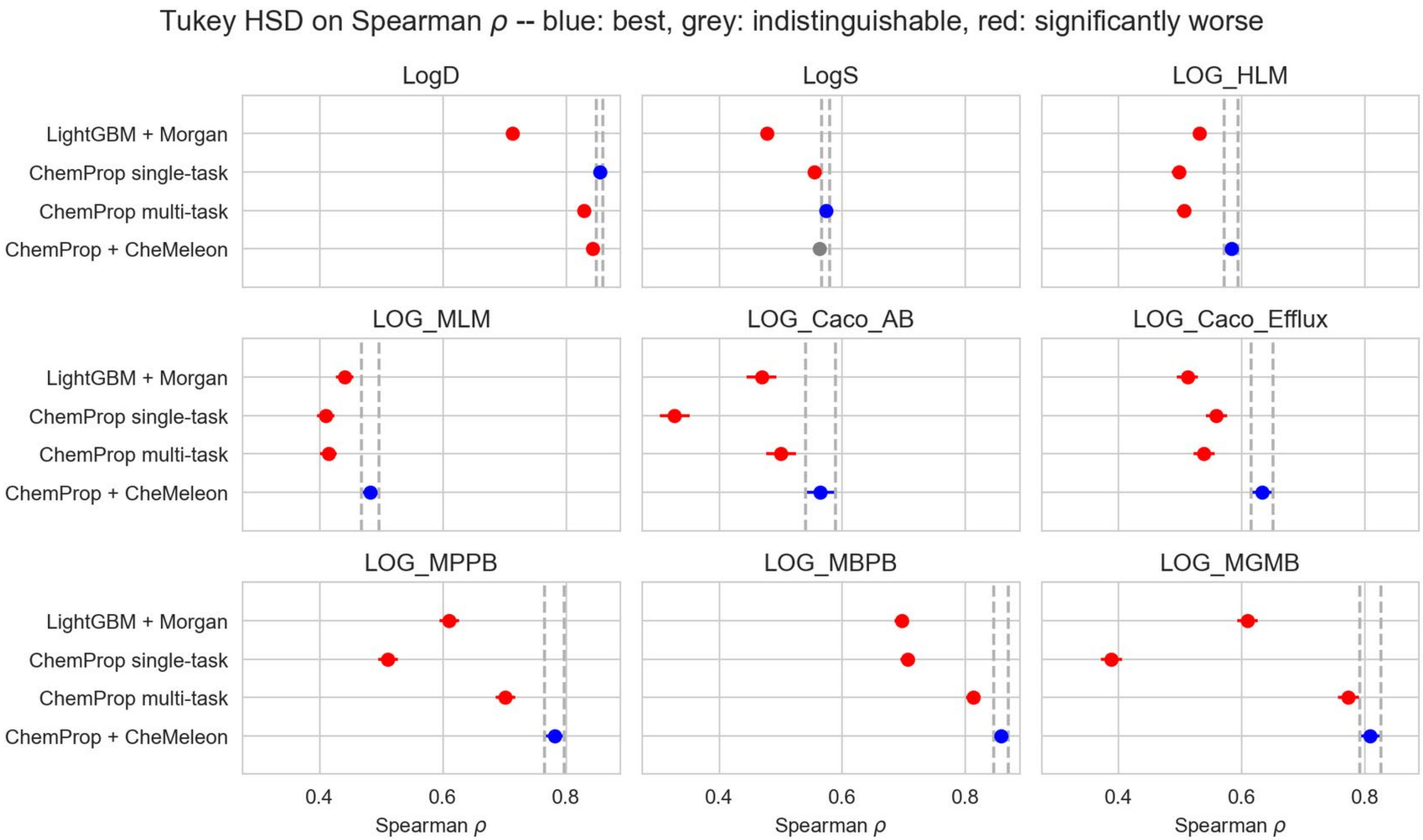
Metric	Formula	Measures	Interpretation	When to Use
MAE	$\frac{1}{N}\sum \mid y_i - \hat{y}_i \mid$	Average absolute prediction error	Lower is better	Interpretability wrt data unit errors, smooth to outliers
RMSE	$\sqrt{\frac{1}{N}\sum (y_i - \hat{y}_i)^2}$	Error with stronger penalty on large mistakes	Lower is better	When large errors are especially costly
R^2	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Improvement over predicting the mean	1 = perfect; 0 = mean predictor; < 0 = worse than mean	Comparing predictive power relative to baseline
Pearson r	$\frac{\mathrm{cov}(y, \hat{y})}{\sigma_y \sigma_{\hat{y}}}$	Linear trend agreement	1 = perfect positive trend	Assessing whether the model learned the trend

Avoid the Dreaded Bold Table

Are these different?

	LightGBM + Morgan	ChemProp single-task	ChemProp multi-task	ChemProp + CheMeleon
LogD	0.713	**0.855**	0.829	0.843
LogS	0.478	0.554	**0.573**	0.562
LOG_HLM	0.531	0.497	0.506	**0.583**
LOG_MLM	0.441	0.410	0.414	**0.482**
LOG_Caco_AB	0.469	0.327	0.501	**0.564**
LOG_Caco_Efflux	0.512	0.559	0.539	**0.633**
LOG_MPPB	0.611	0.511	0.701	**0.782**
LOG_MBPB	0.696	0.706	0.813	**0.857**
LOG_MGMB	0.609	0.388	0.773	**0.808**

Comparing Performance:
Tree-Based Method, Graph Neural Network, Foundation Model

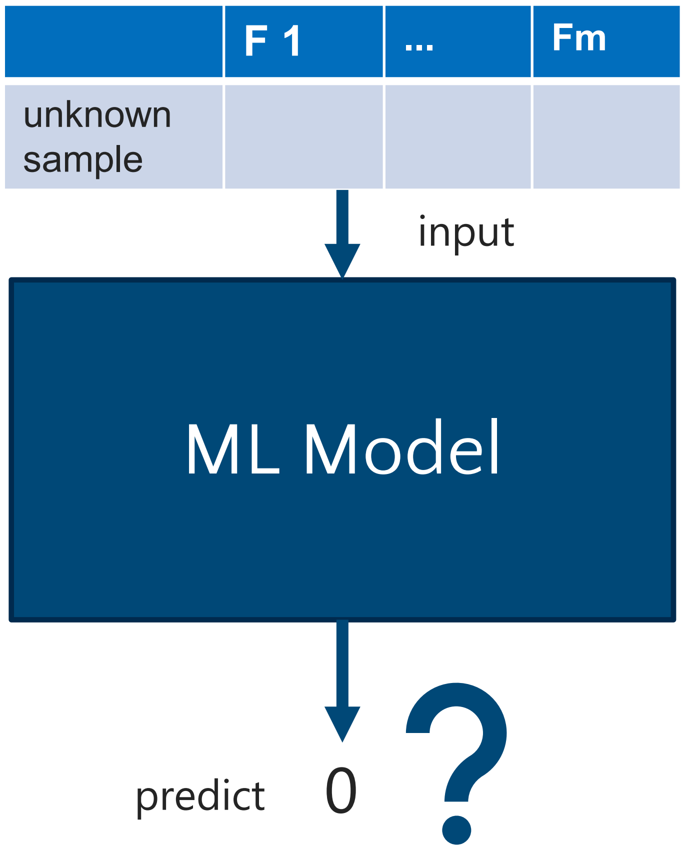


Increasing trust through

Reliability

Does the model work in domain?

Model Uncertainty



We know that

- the model is properly trained using the training data
- the model reached an MCC of 0.9 on the test data

How well does it predict a new sample?

Performance measures indicate how well the model performs overall,

but **is a specific prediction likely to be correct or not?**

➢ Model *(un)certainty* for a specific prediction

Epistemic and Aleatoric Source of Uncertainty

Assumption in supervised ML

- Functional relationship approximated between features and response
- But radomness/noise in the data

epistemic uncertainty (reducible)

- model uncertainty (hypothesis space)
- approximation uncertainty (model parameters)

approximation of real functional relationship

aleatoric uncertainty (irreducible)

irreducible error (data)

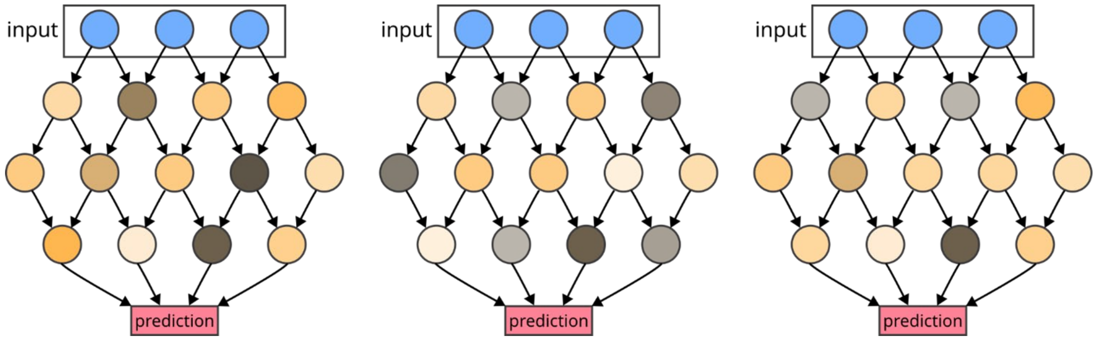
response vector

$$y = \hat{f}(X) + \varepsilon$$

feature matrix

Using Uncertainty to Assess Model Confidence

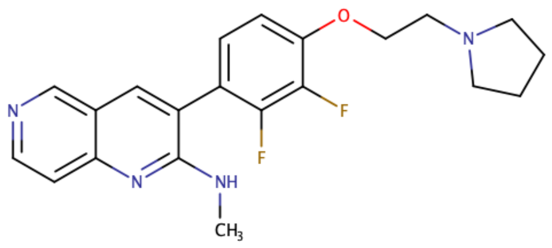
- **Single deterministic methods**
  - Classification setting: predicted class probabilities
  - Secondary model trained to predict uncertainty for an already trained model
- **Ensemble methods**
  - Ensemble of similar but different models -> Variance as uncertainty
  - Varying random seeds, resulting in different weights
  - Varying the training data (*bagging*)
- **Training- and test-time data augmentation**
  - Augmeted set of points using stochastic noise or domain knowledge



https://github.com/volkamerlab/teachopencadd

Ensemble Variance as an Uncertainty Estimate: Random Seed

- Create multiple models with different random seed
- More confidence when values are predicted consistently

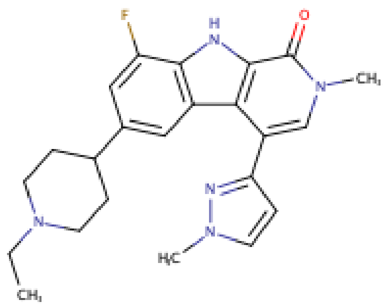


Test set molecule 1

Model	Prediction
1	5.1
2	3.6
3	7.1
4	4.2
5	5.0

σ= 1.19

High prediction variance
Low confidence



Test set molecule 2

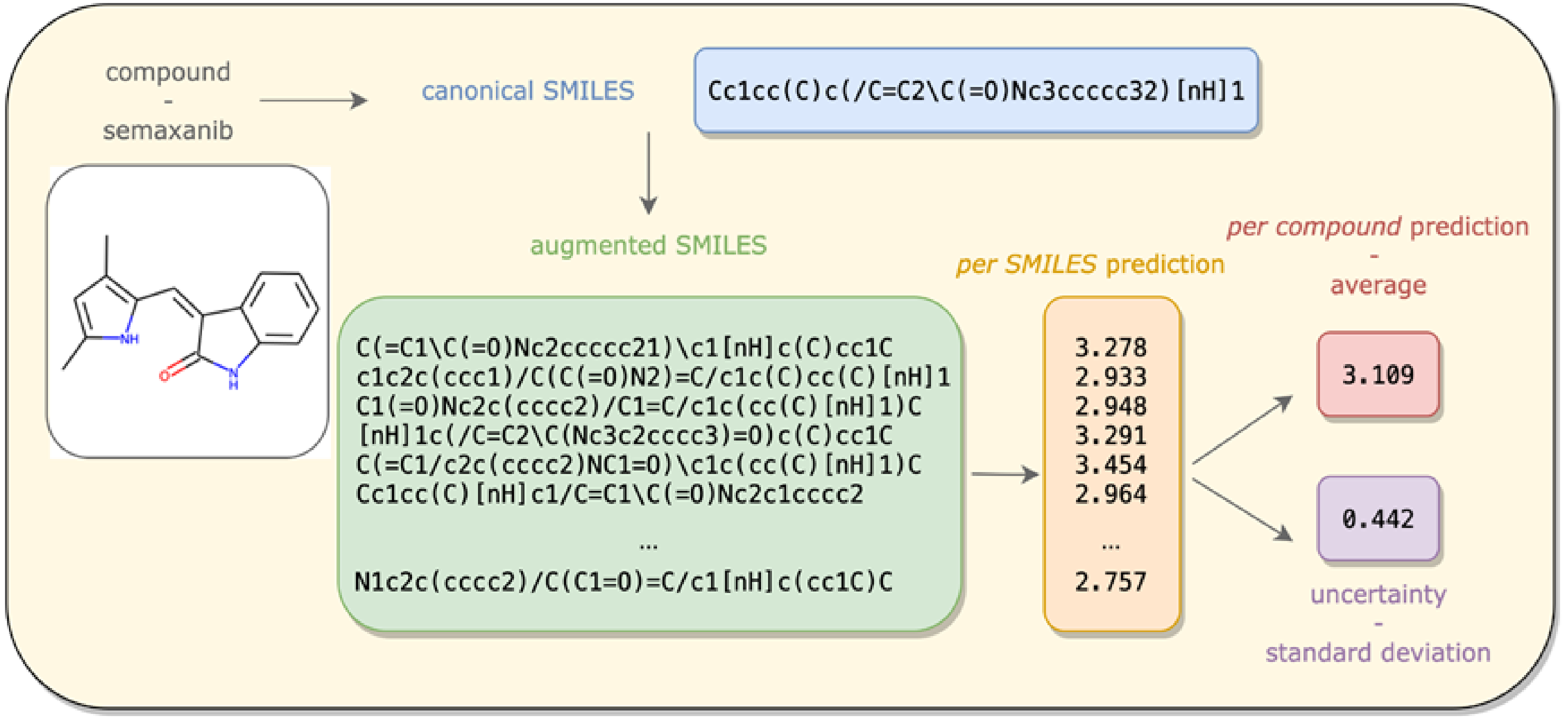
Model	Prediction
1	5.1
2	5.2
3	5.1
4	5.0
5	5.3

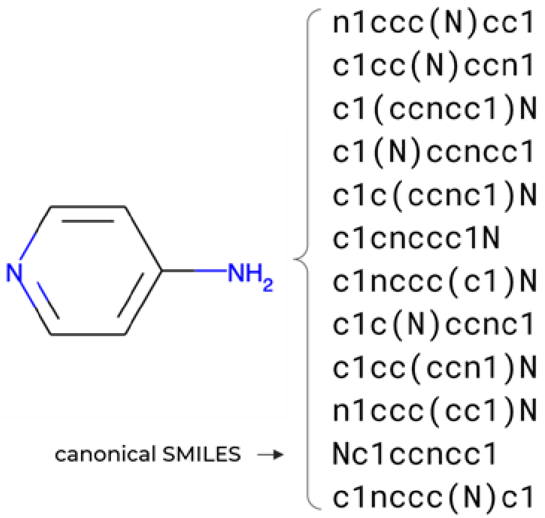
σ= 0.10

Low prediction variance
Hight confidence

Ensemble Variance as an Uncertainty Estimate: Data Augmentation

SMILES augmentation improves model performance





Kimber *et al.*, *AI in Life Sciences*, **2021**, 1, 100014
https://github.com/volkamerlab/maxsmi

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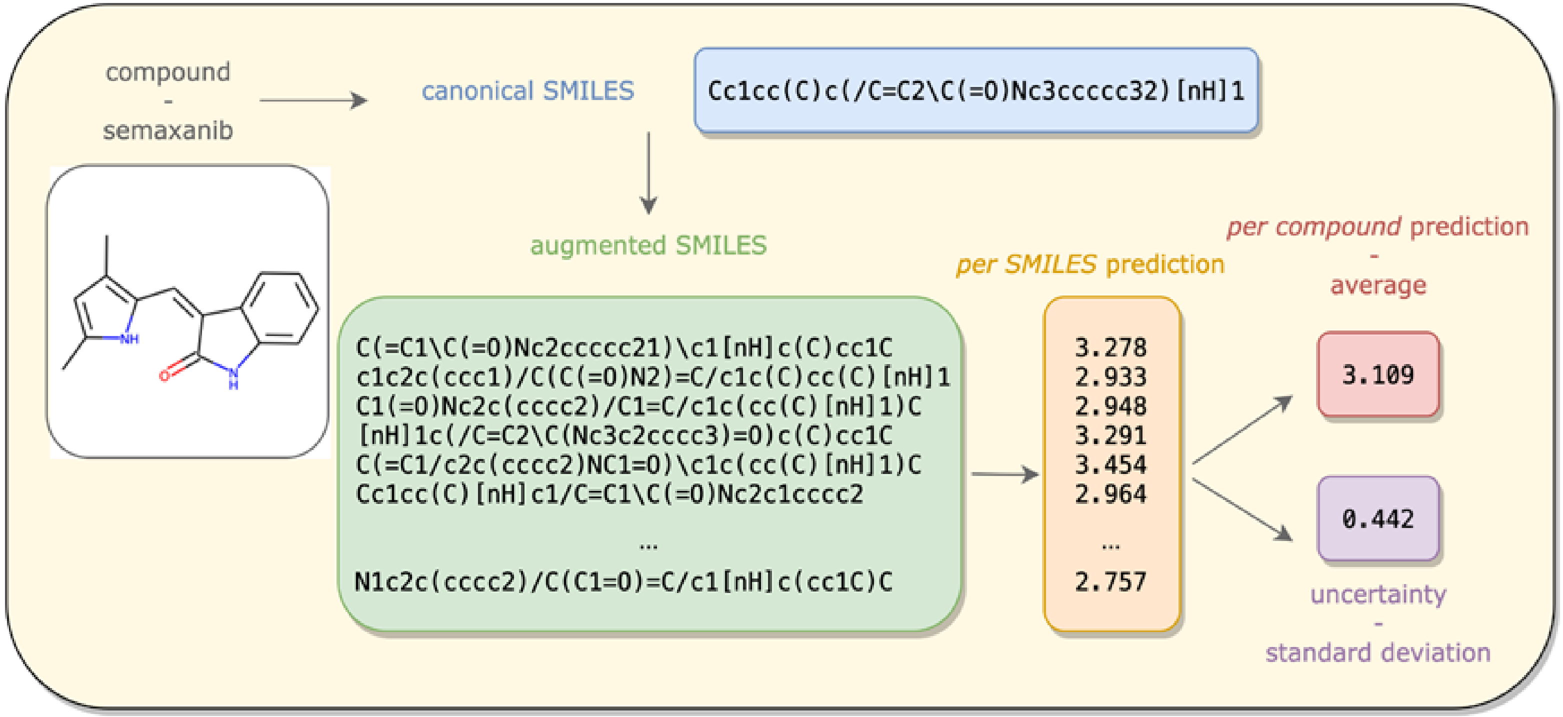

Kimber *et al.*, *AI in Life Sciences*, **2021**, 1, 100014

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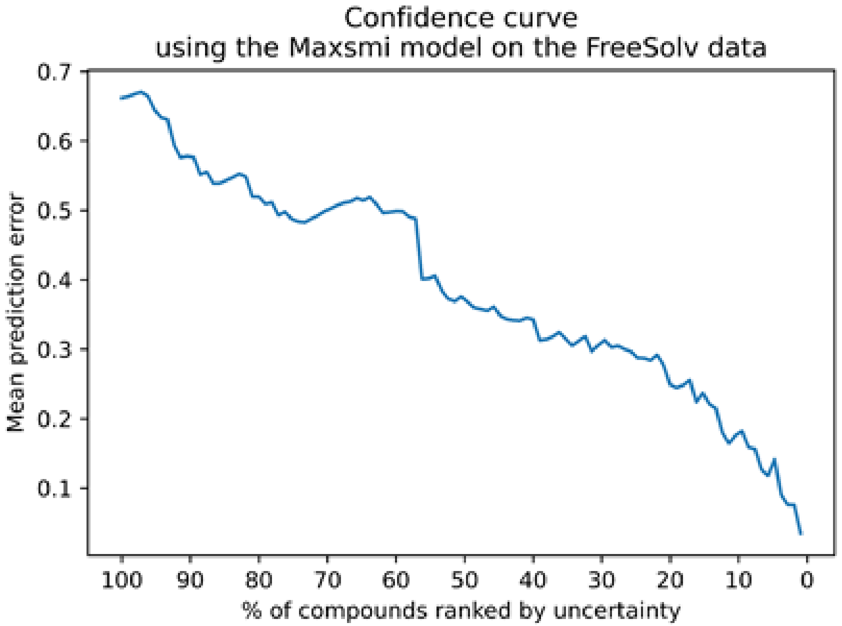
Uncertainty prediction

More confident model implies smaller prediction error



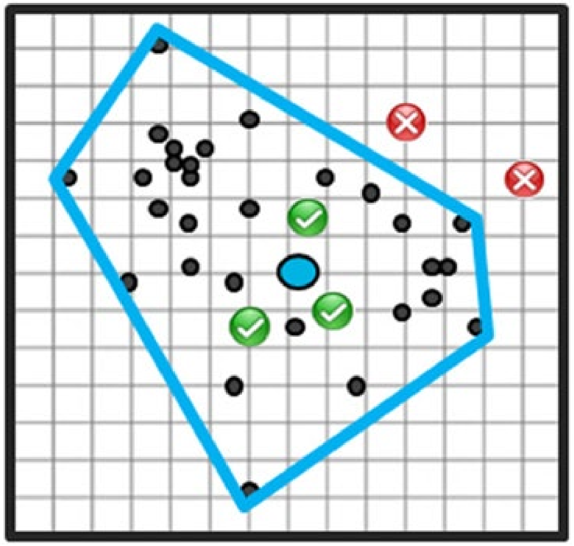
Increasing trust through

Applicability Domain

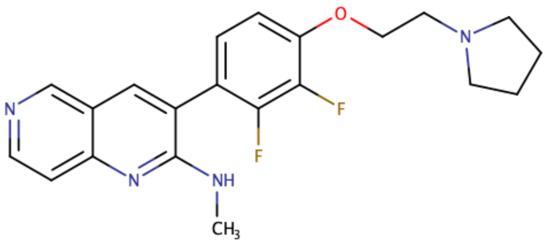
Model Applicability: Distance to Training Set as a Uncertainty Metric

Simple: Express model applicability as distance to the 5 closest training set compounds

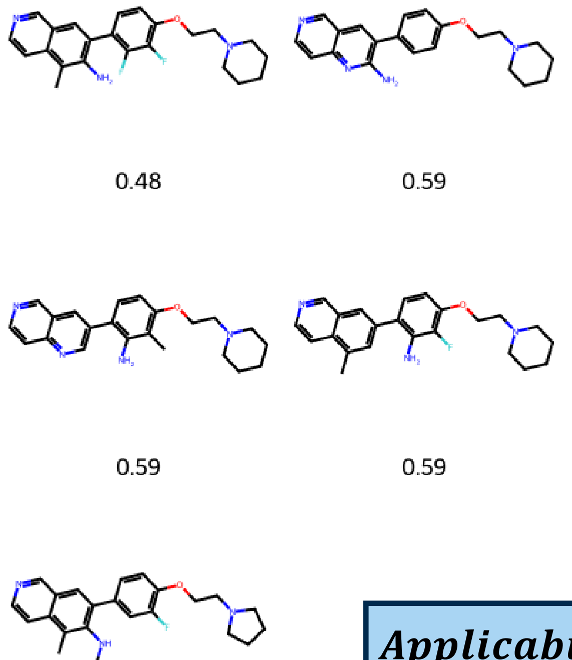
Interpolation within or extrapolation from the chemical structure space



Test set molecule



Most similar training set molecules



0.48 0.59

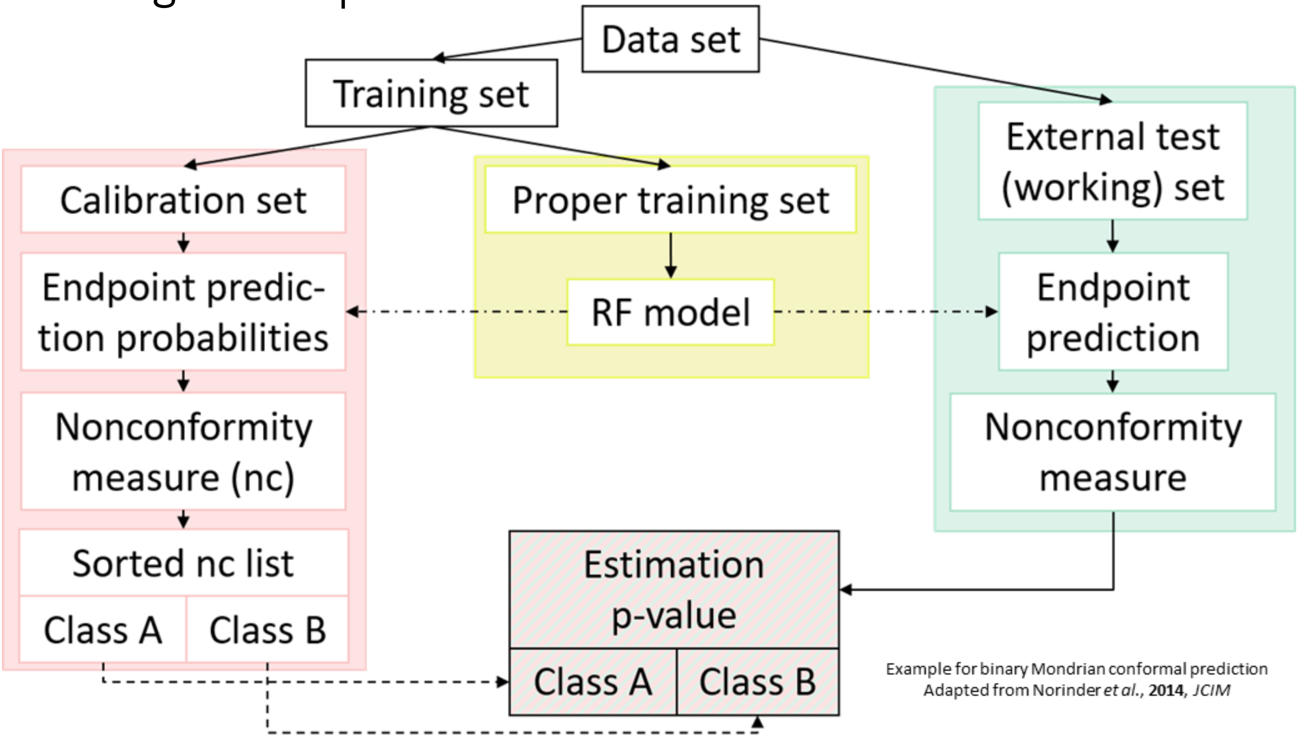
0.59 0.59

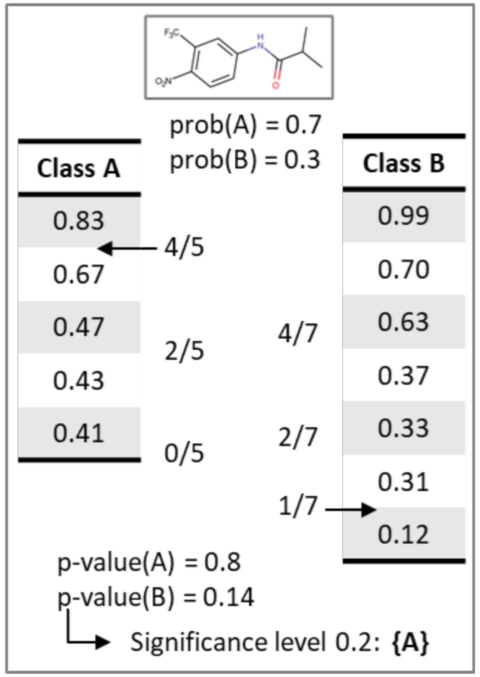
0.59

Applicability score $= \frac{\mathbf{0.48+0.59+0.59+0.59+0.59}}{\mathbf{5}}$ = 0.57

Conformal Prediction Framework Based on ML

- Statistically valid on a given confidence level (if exchangeability is given)
- Additional calibration step: Compare predictions to those previously seen (p-value)
- Output prediction sets in binary classification: **{A}, {B}, {A,B}** or **{ }**
- Validity = % of correct classifications
- Efficiency = % of single class predictions





Example for binary Mondrian conformal prediction
Adapted from Norinder *et al.*, **2014**, *JCIM*

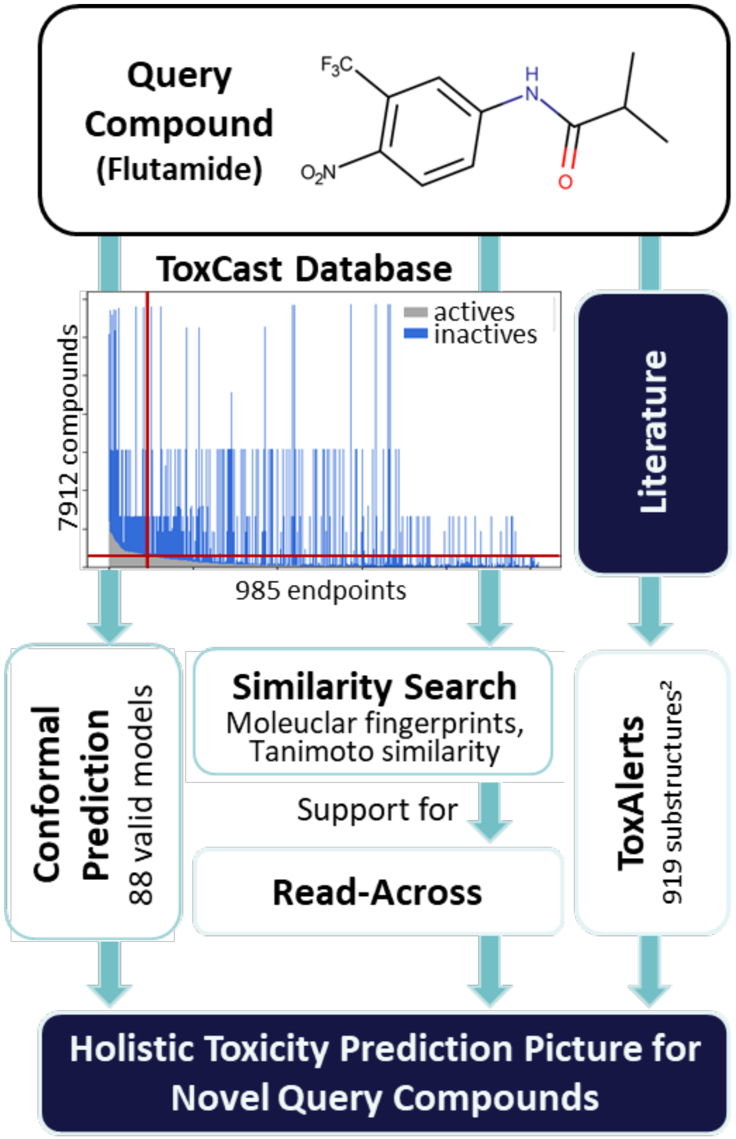
Know-Tox – Predictions with Confidence

- ToxCast data set
  - ~ 8000 compounds * 1000 endpoints (sparse)
  - Pharmaceutical, pesticides, environmental chemicals
  - Cell cycle, steroid receptors, cytotoxicity, …

Can we apply the model to new (BASF) data?

- Conformal prediction
  - Case study: Antiandrogen activity (AA)
  - Validity, efficiency and accuracy like other studies

	Dataset	Efficiency	Accuracy (SCPs) all	cl.1	cl.0	# toxic/ non-toxic
cross-validation	ToxCast AA	0.87	0.78	0.80	0.78	868/5842
external	Norinder[3]	0.79	0.68	0.70	0.67	160/201
In-house	BASF (I)	0.94	**0.56**	**0.97**	**0.07**	280/254



Know-Tox – Predictions with Confidence

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  - Cell cycle, steroid receptors, cytotoxicity, …

Can we apply the model to new (BASF) data?

- Conformal prediction
  - Case study: Antiandrogen activity (AA)
  - Validity, efficiency and accuracy like other studies
  - kNN normalization & balancing
  - Less efficient, but higher accuracy

	Dataset	Efficiency	Accuracy (SCPs) all	Accuracy (SCPs) cl.1	Accuracy (SCPs) cl.0	# toxic/ non-toxic
cross-validation	ToxCast AA	0.87	0.78	0.80	0.78	868/5842
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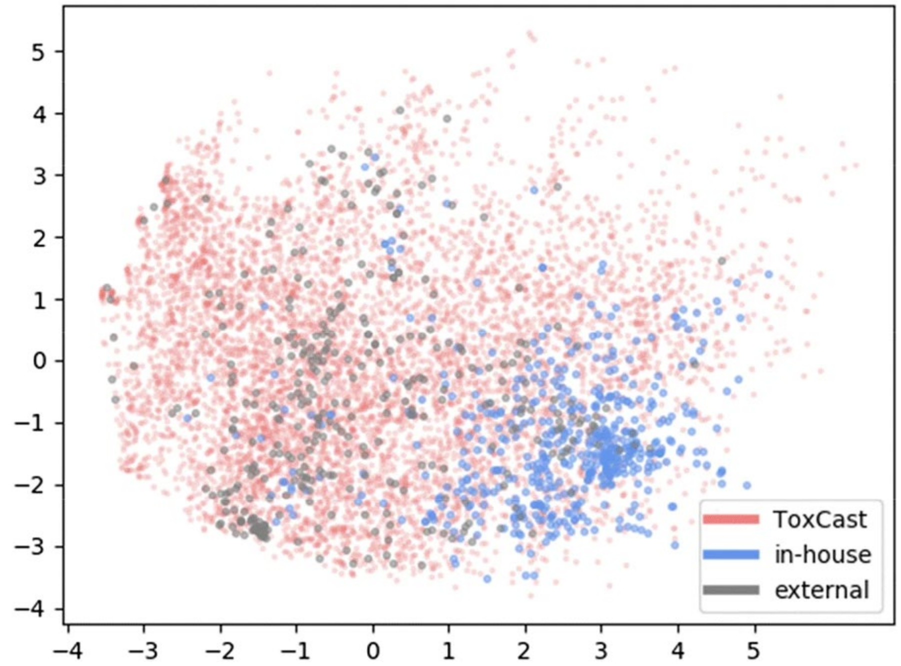
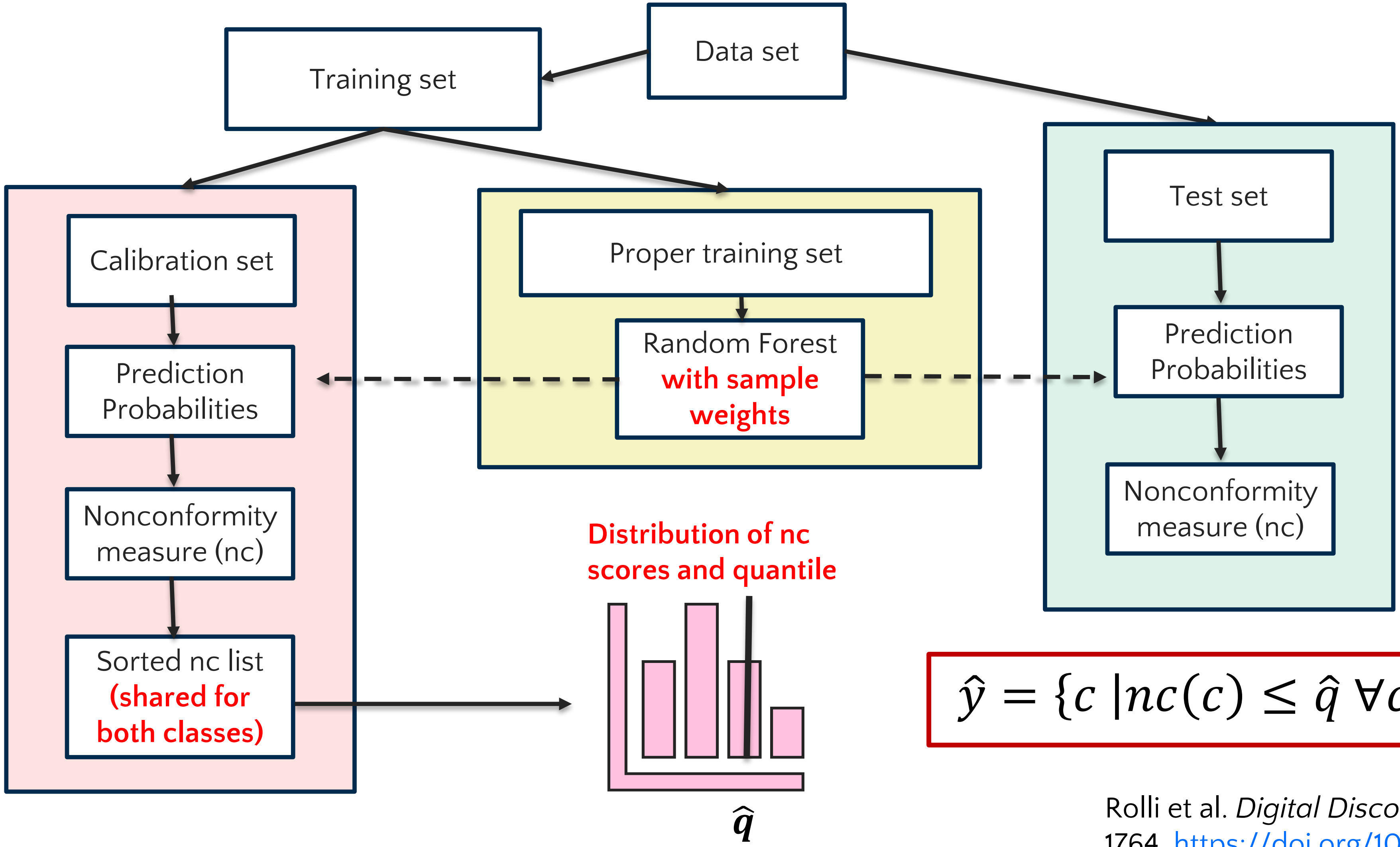
in-house	BASF (I)	0.94	**0.56**	**0.97**	0.07	280/254
	BASF (II)	**0.20**	**0.75**	**0.80**	**0.71**	280/254



Descriptor space of a 2-component PCA trained on ToxCast-AA data

Morger, A., *et al. J Cheminform* 12, 24 (**2020**)
https://github.com/volkamerlab/knowtox_manuscript_SI

Conformal Prediction with Sample Weights to Account for Class Imbalance



$$\hat{y} = \{c \mid nc(c) \leq \hat{q} \; \forall c\}$$

Rolli et al. *Digital Discovery* 2026; 5 (4): 1746–1764. https://doi.org/10.1039/d5dd00284b
https://github.com/volkamerlab/MORGOTH

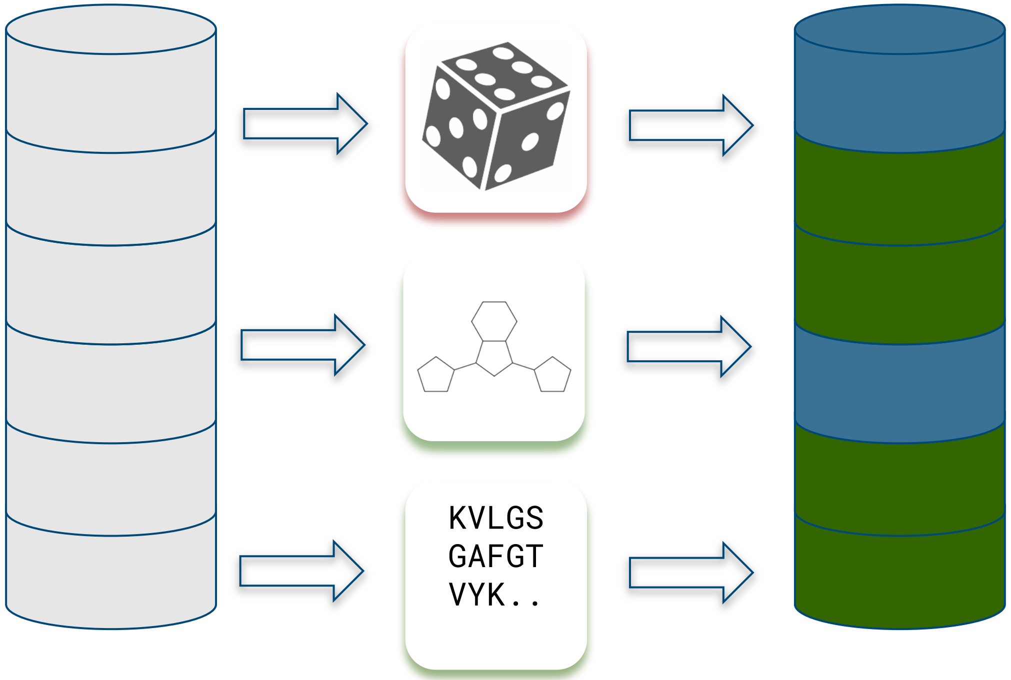
Increasing trust through

Robustness

Does the model work outside domain?

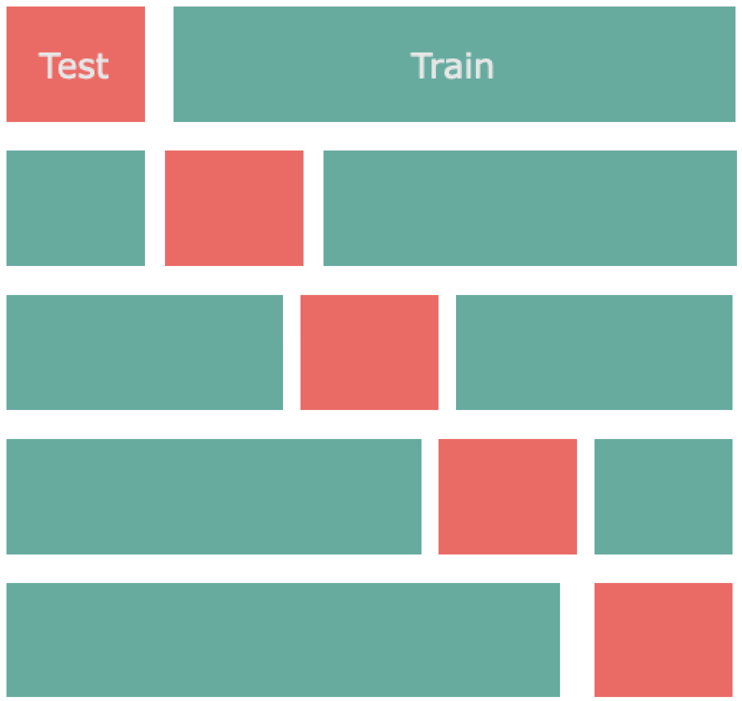
Data Split is Crucial to Assess Generalizability

- Train/val/test split
- Random splits overestimate out-of-domain generalization
- Better: Splits based on
  - Ligand scaffold
  - Assay
  - Protein sequence

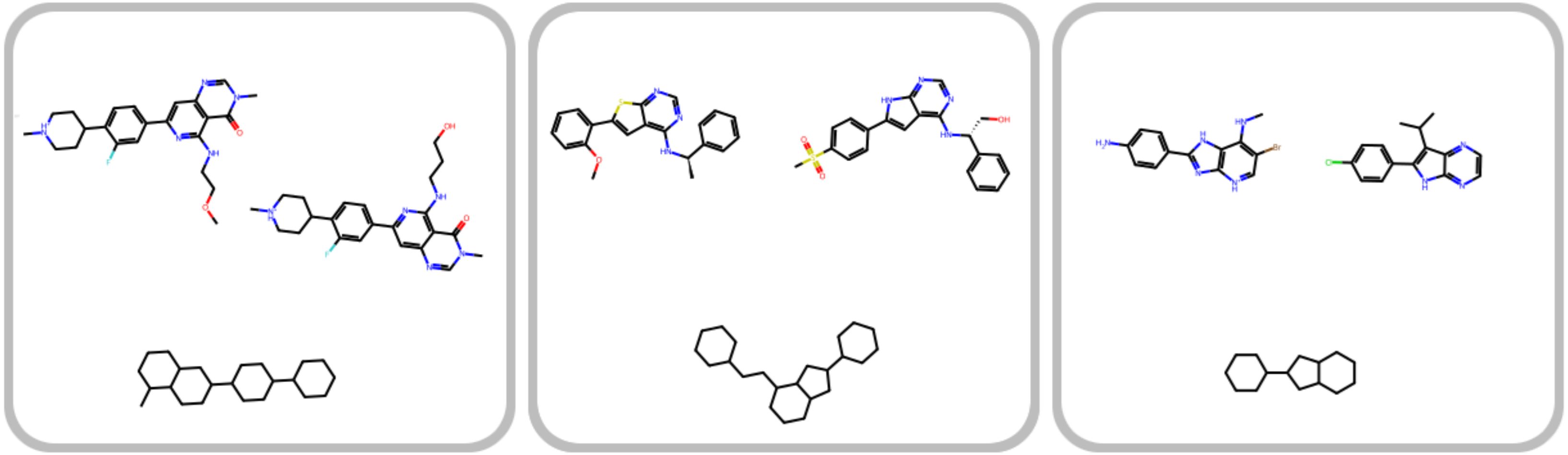


Grouped k-Fold Split

- Folds are in accordance with groups (scaffold clusters)
- Every data point is part of the test set *exactly once*
- Folds may not be of exact equal size
- Consider labels (stratified split)



Data Splits – Scaffold Split



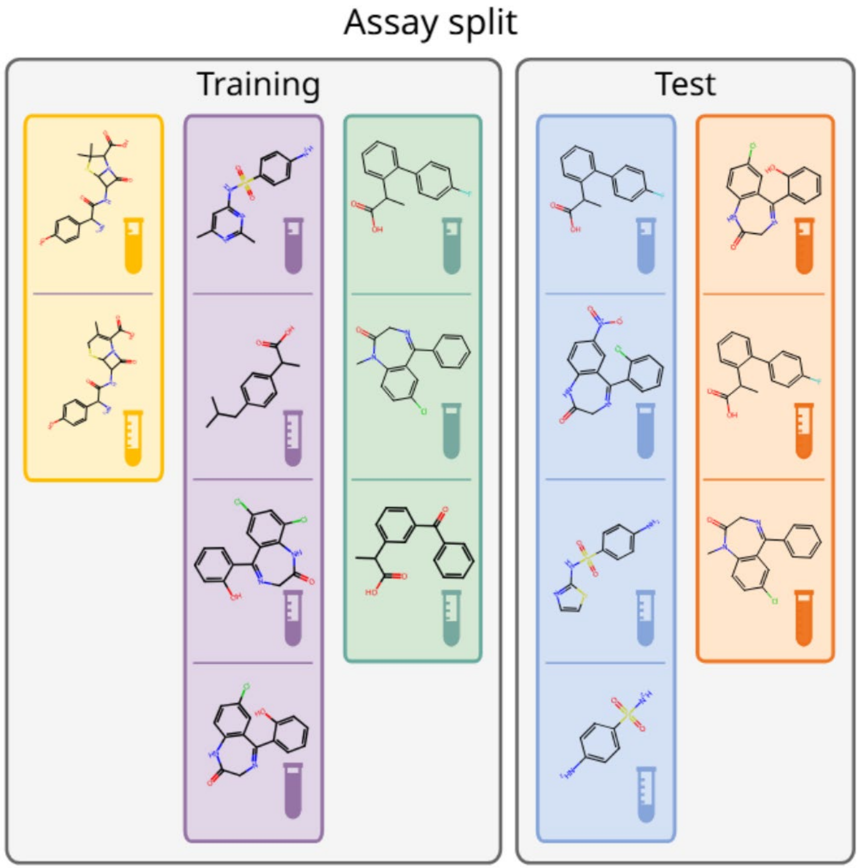
- Avoid data leakage over structurally similar ligands
- Group by generic scaffold *(or similarity)*
- Ensure each group only in one of {train, test, val}

Data Splits - Assay-Based Split

- Molecules tested within the same assay tend to be structurally related and often exhibit similar activity patterns due to shared experimental protocols

- Partition the data based on the assay ID

- Let's us see what we can learn from one set of experiments for another set of experiments



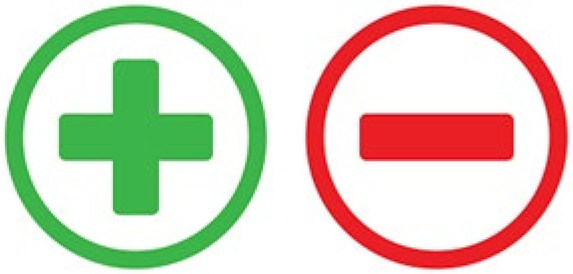
Backenköhler M., Groß J., Volkamer A.. Assay-Based Machine Learning: Rethinking Evaluation in Drug Discovery. *ChemRxiv*. 23 February 2026. https://doi.org/10.26434/chemrxiv-2025-zsk6d/v2

DataSAIL - Data Splitting Against Information Leakage

Developed by Roman Joeres at Prof. Kalinina’s lab, Saarland University
uses mathematical optimization to identify the most difficult split
https://github.com/kalininalab/DataSAIL

Take Home Messages

- More data available
- Increased computational power
- Advances in neural network algorithms
- Open-source libraries



- True predictive power
- Applicability and reliability
- Interpretability

- **Data quality, relevance and proper processing are key**
- **Proper model evaluation, always consider baselines when applying ML/DL**
- **Include trustworthiness considerations in your evaluation**

Github with more teaching material
https://github.com/volkamerlab/TeachOpenCADD

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Thank you for your attention!

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